AAYTES_01 project

scRNAseq análisis

6 samples merged

6 mouse prostate samples

Fastq documents

10x single cell technology and sequenciation RNA-seq

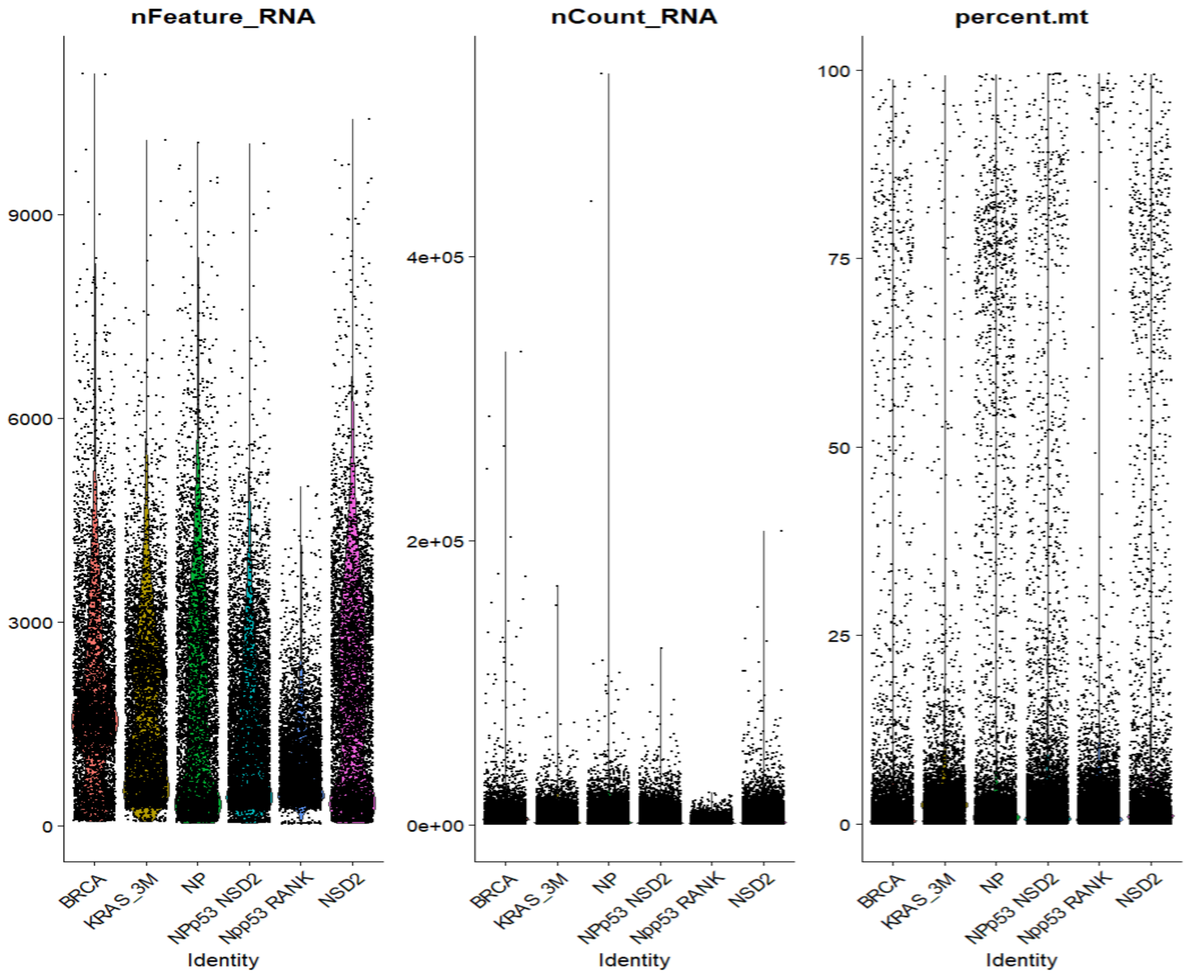
Counts(H5 matrix)

Cell ranger counts: Tool that convert the sequences in to counts of genes expressed using a mouse reference

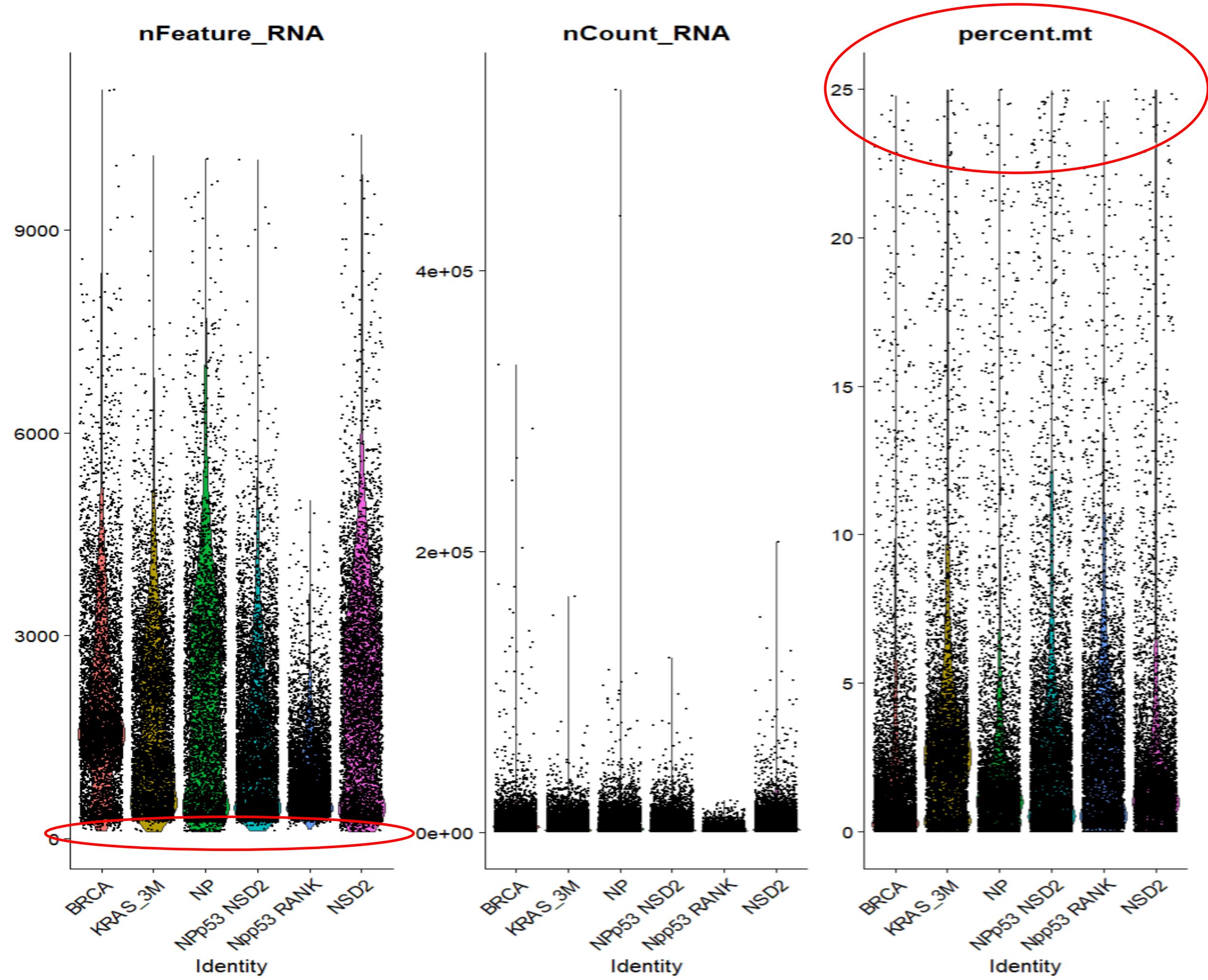
R:
Data integration and normalization
Seurat single cell analysis
Graphs

Single cell reports: PDF with graphs and explanations, csv with the gene markers of every cluster.

Quality control

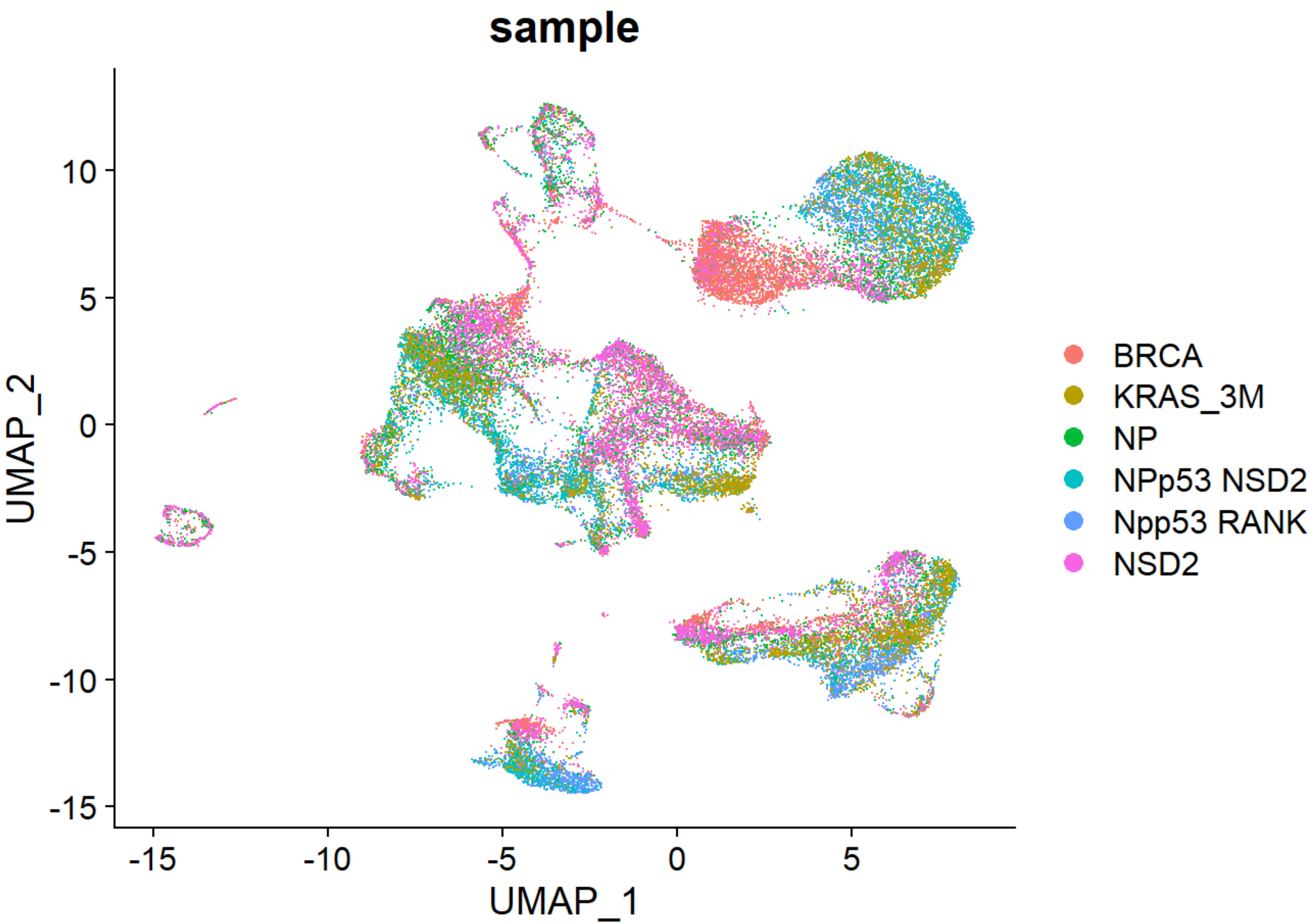


Violin plot of raw data. For every sample we can see the distribution of the values of the variables nFeatures(number of genes by cell), nCount(number of count by cell) and percent.mt which is the mitocondrial RNA percetange found by cell



After filtering by number of genes by cell(nFeatures) higher than 100 and by mitocondrial percentage lower than 25

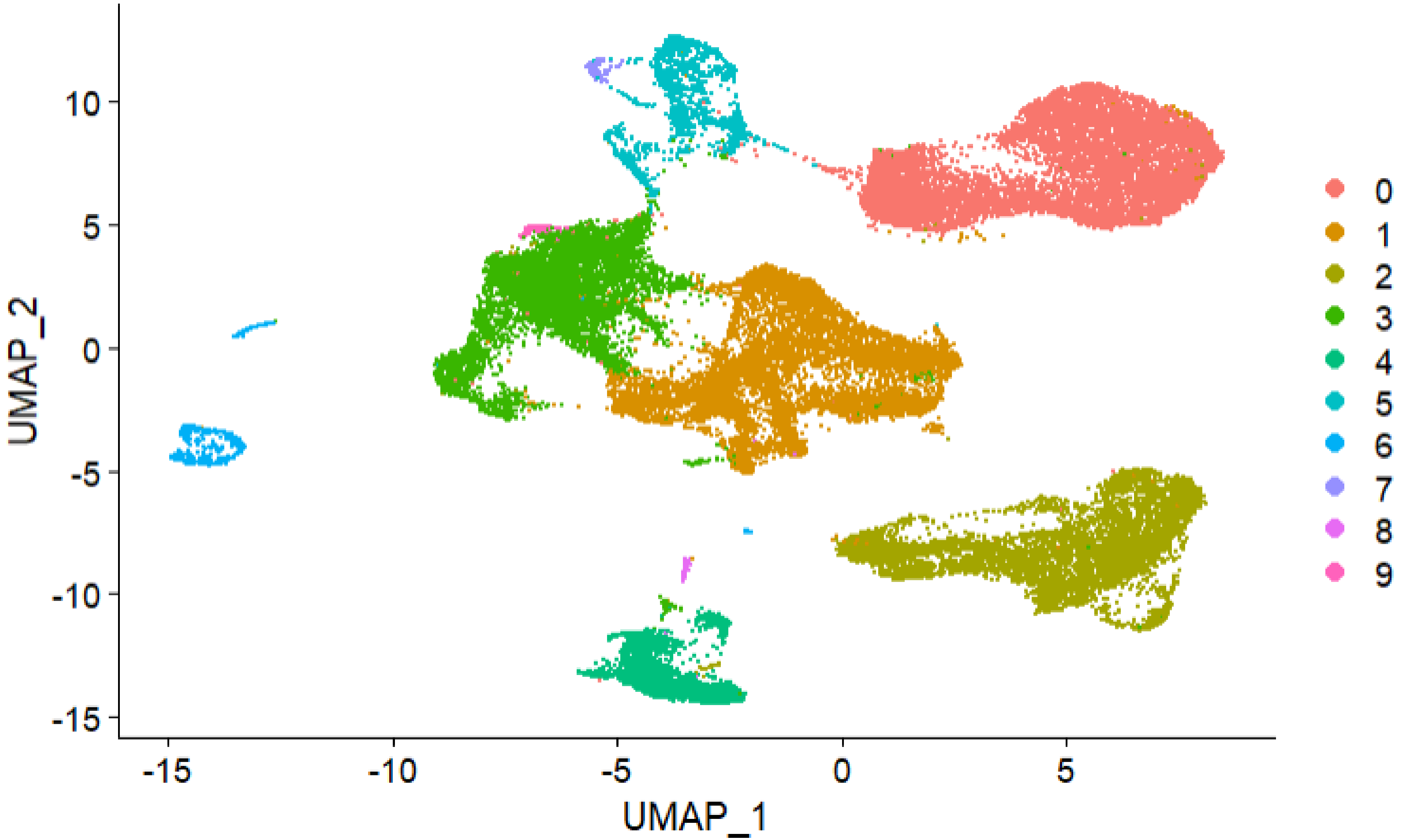
Umap group by sample



In this graph(UMAP) every point is a cell and each colour is a different samples. The position in the graph it's depend on the expression of the cell, then all the similar cell should cluster together.

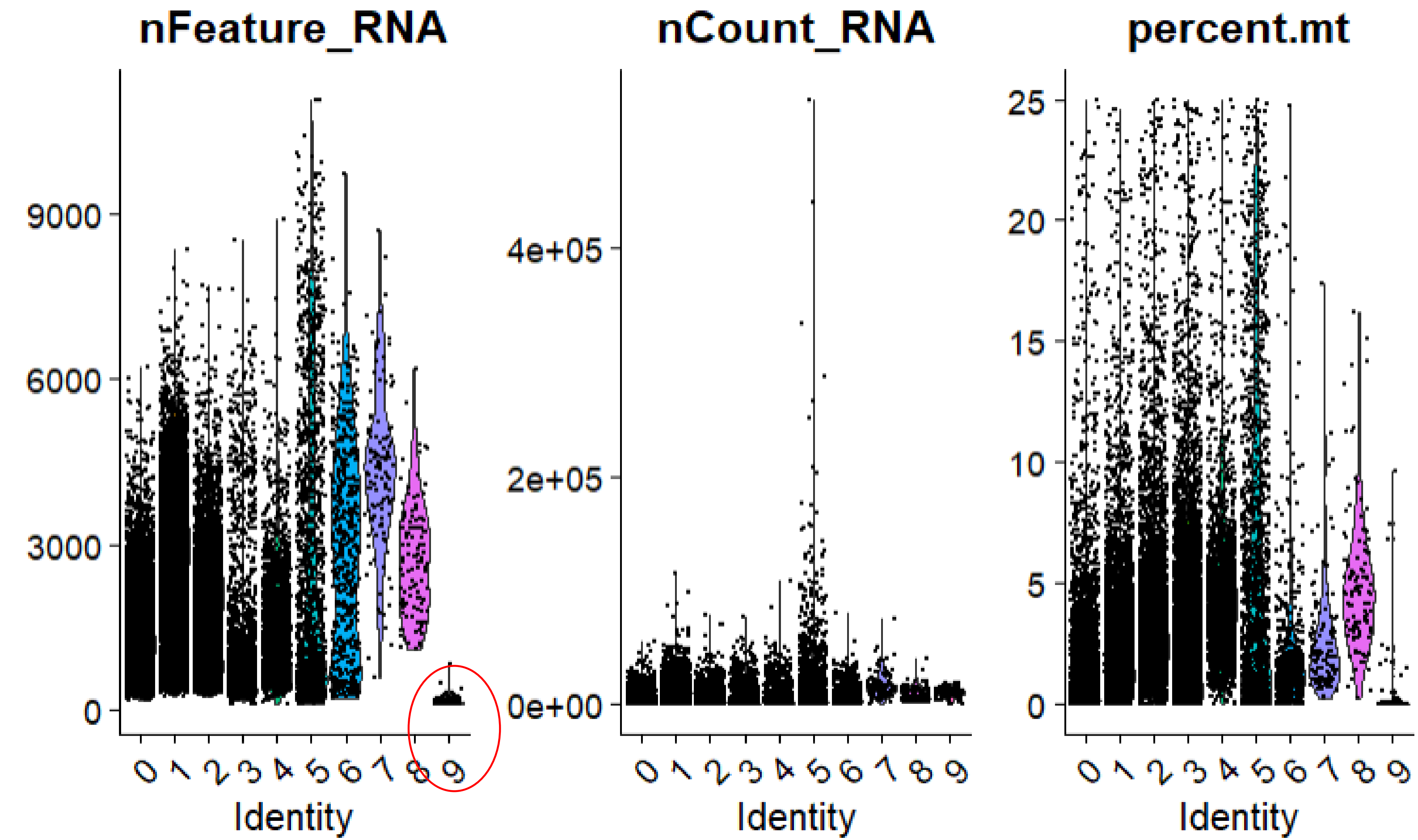
Here we can see the 6 samples integration it's very good, the population of the samples are similar, which is a good indicator of the quality.

Umap group by cell type cluster



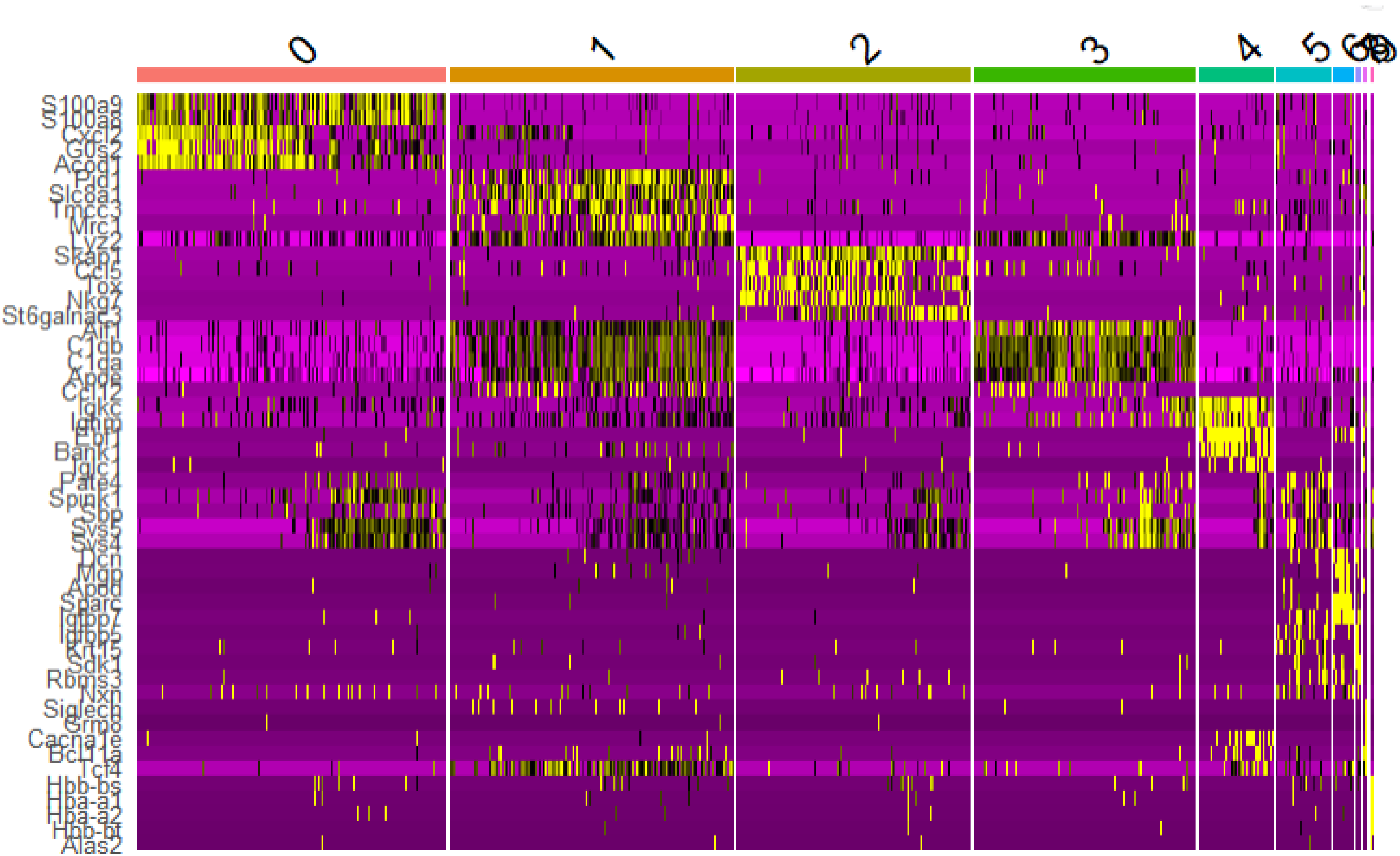
In this UMAP we calculate and represent the clustering of the cells. Each group should represent a different cellular type or subtype.
We found 10 cluster at this level of resolution(0.1).

ViolinPlot group by clusters



Now we represent the features, counts and mitochondrial.percent by cluster. We can see the cluster 9(red circle) has very few genes by cells, it must be a technical noise. All the rest cluster looks good enough.

Umap 5 most difference expressed genes between clusters

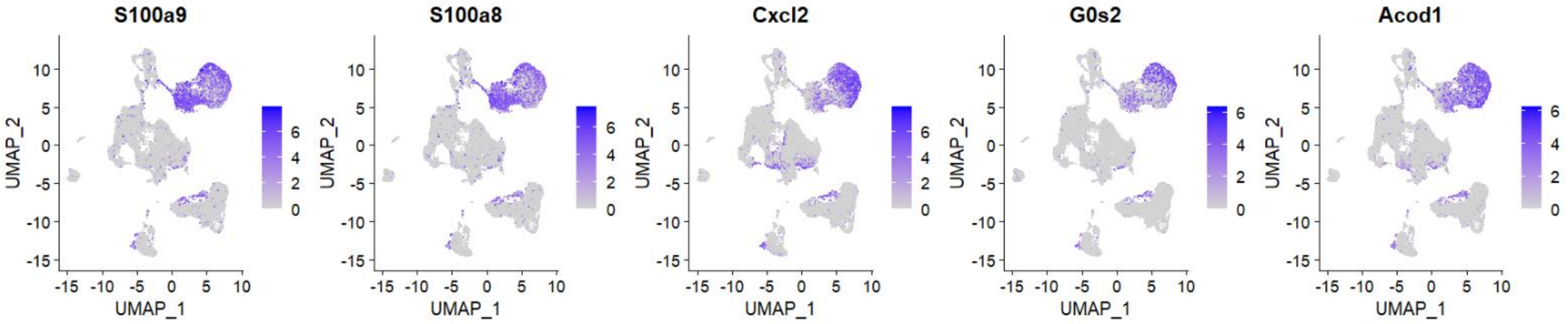


In this heatmap we can see the top 5 genes differentially expressed by cluster.

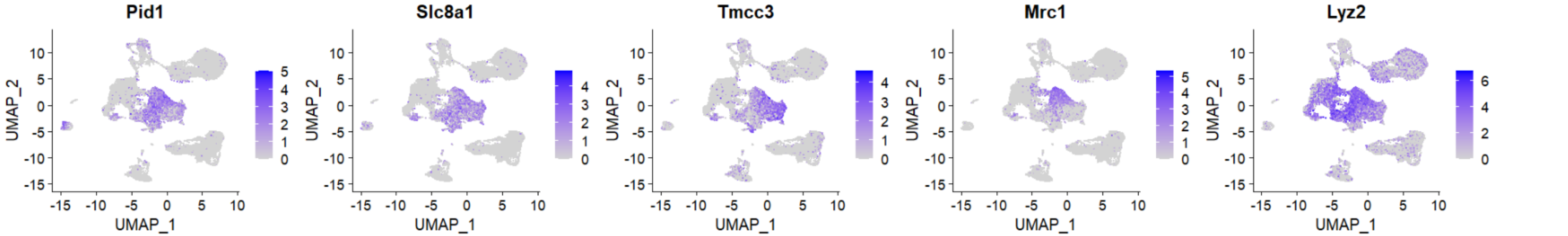
In the next slides we can see the expression of this genes in the UMAP representation.

Feature plot of top5 markers

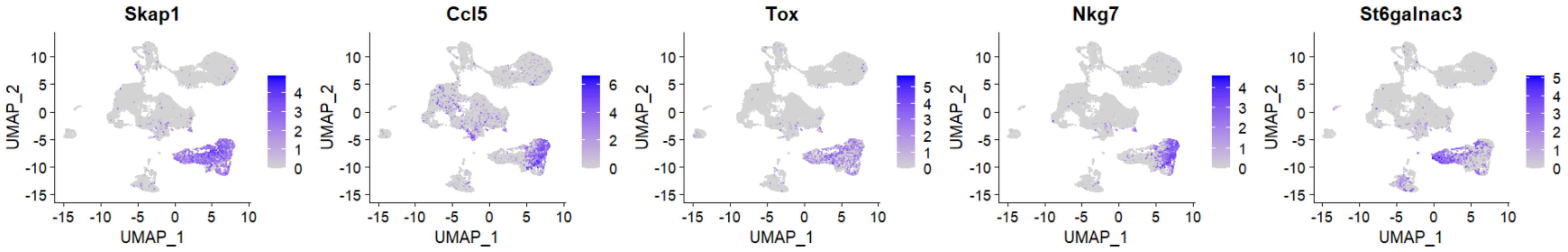
Cluster0



Cluster1

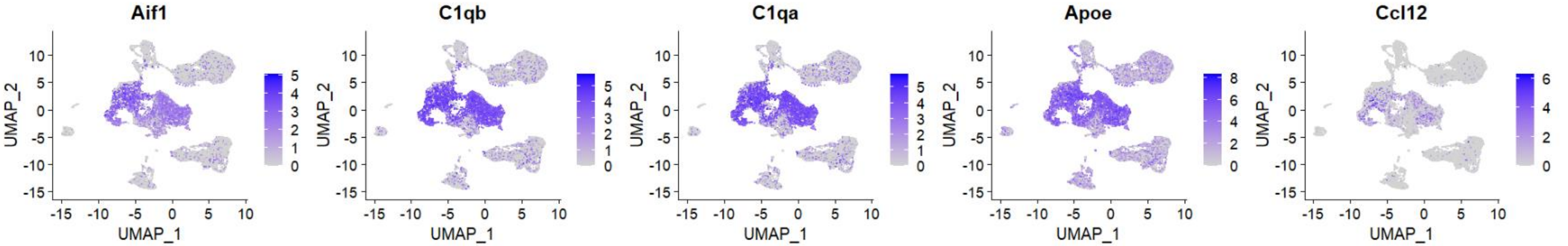


Cluster2

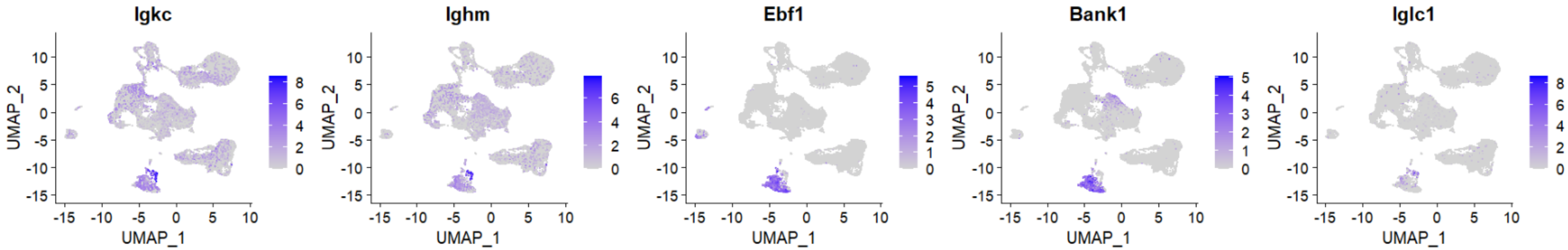


Feature plot of top5 markers

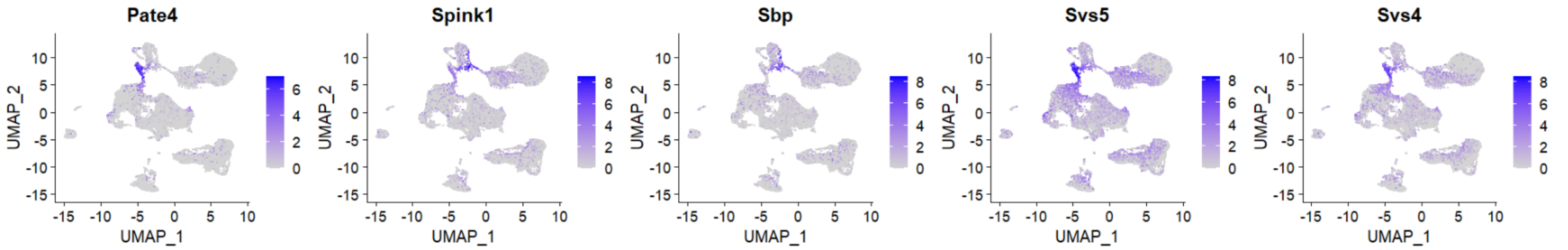
Cluster3



Cluster4

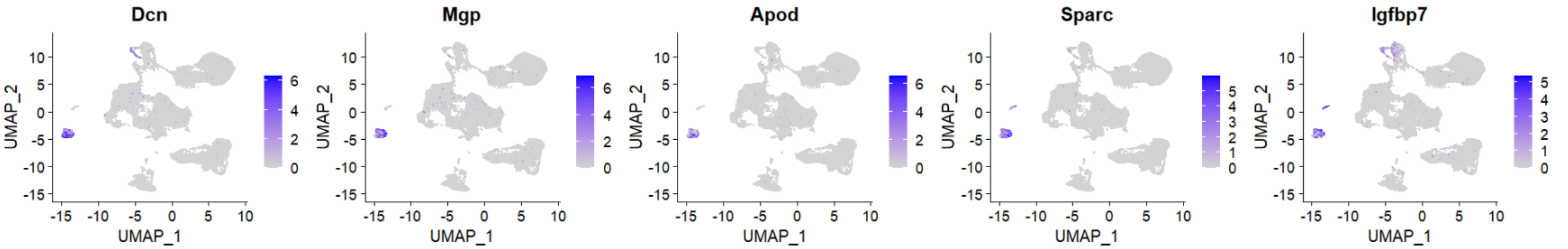


Cluster5

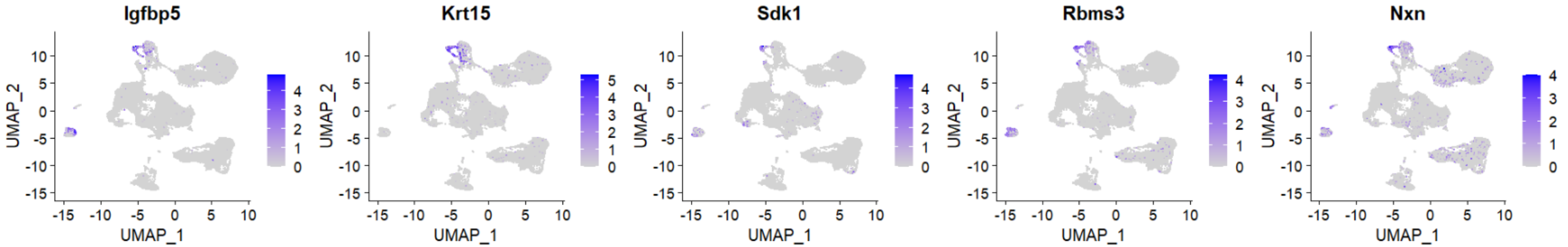


Feature plot of top5 markers

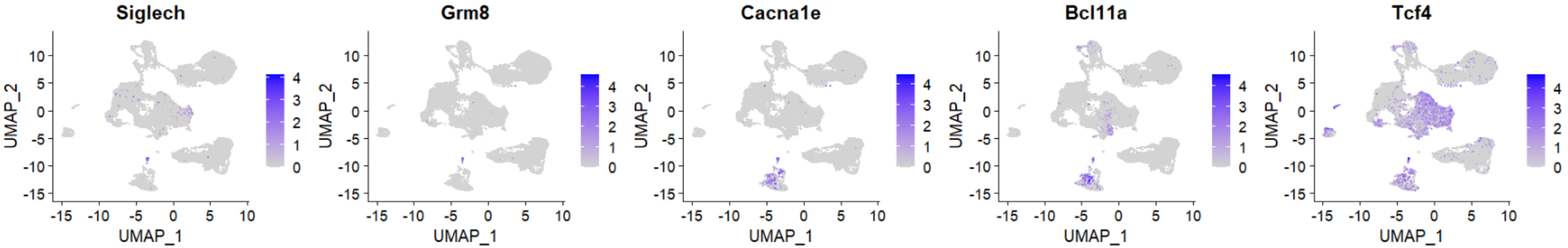
Cluster6



Cluster7

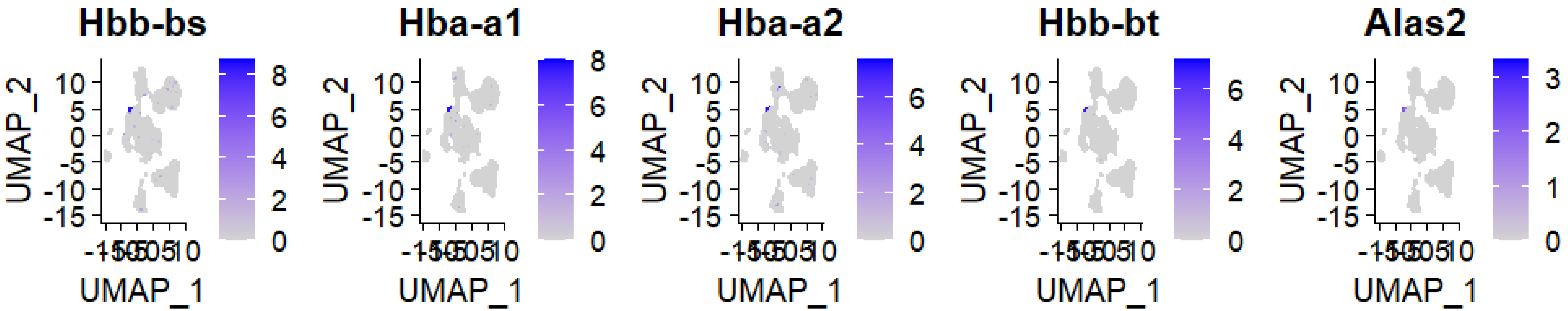


Cluster8



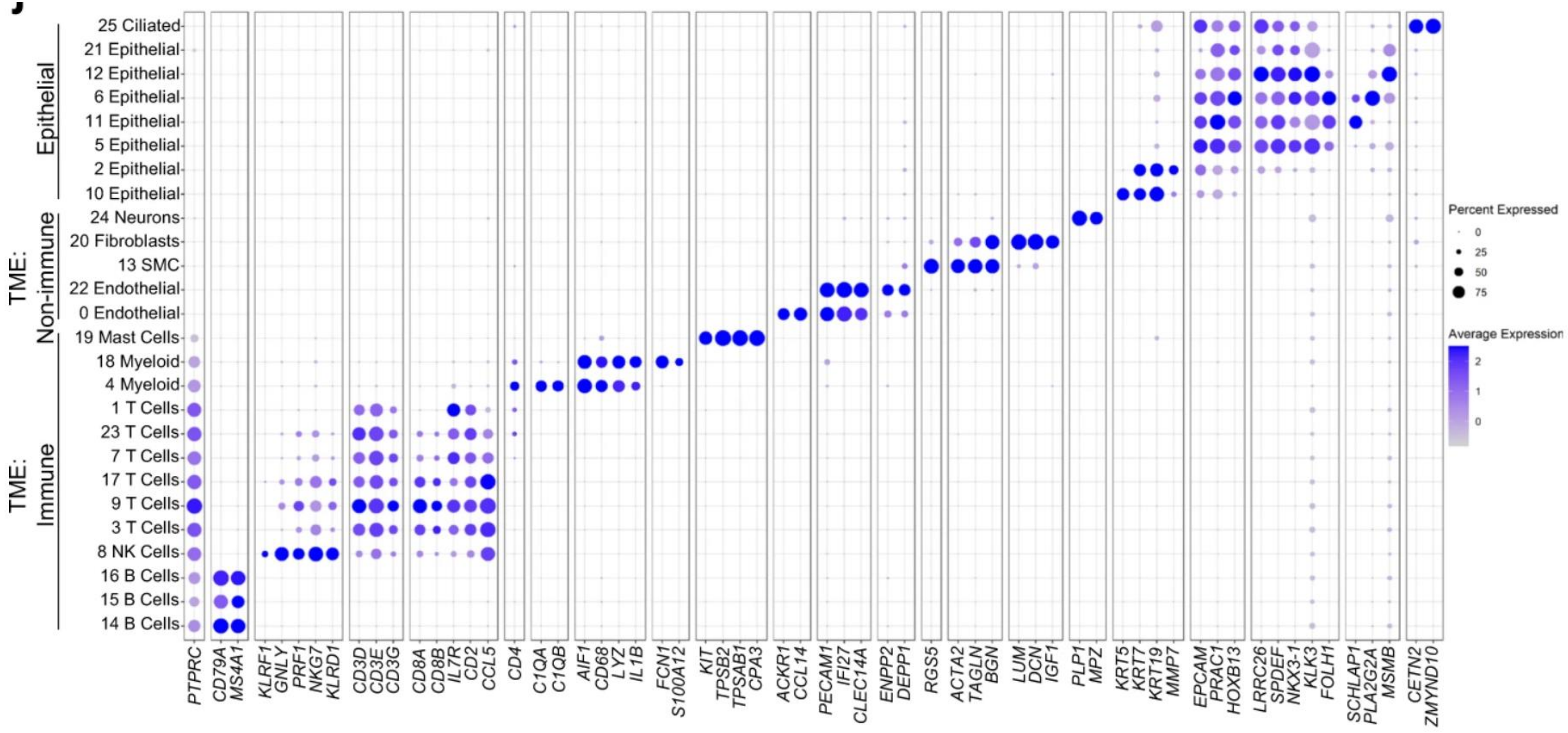
Feature plot of top5 markers

Cluster9

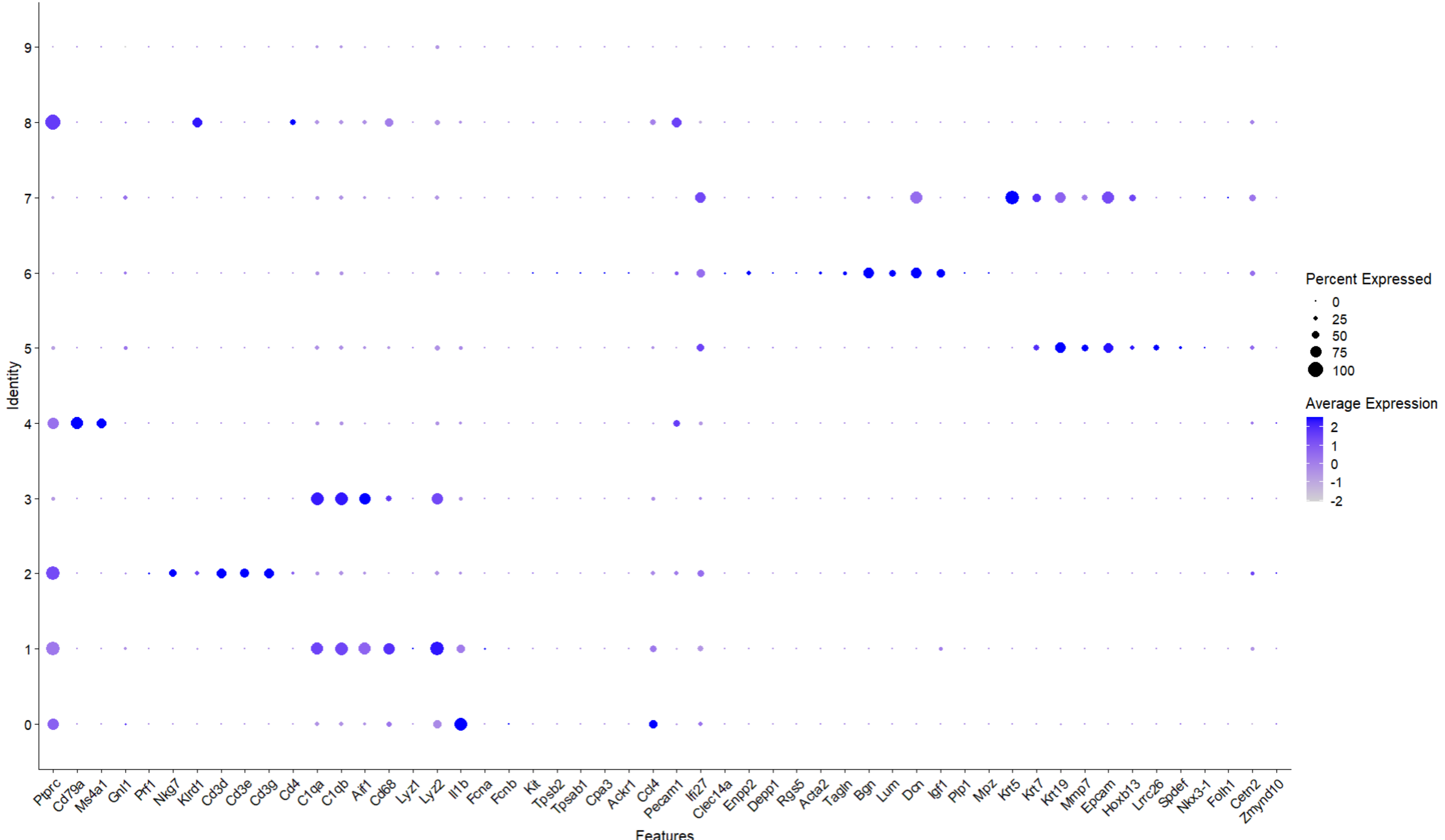


To kwon cell type identity of every cluster, we must use a reference

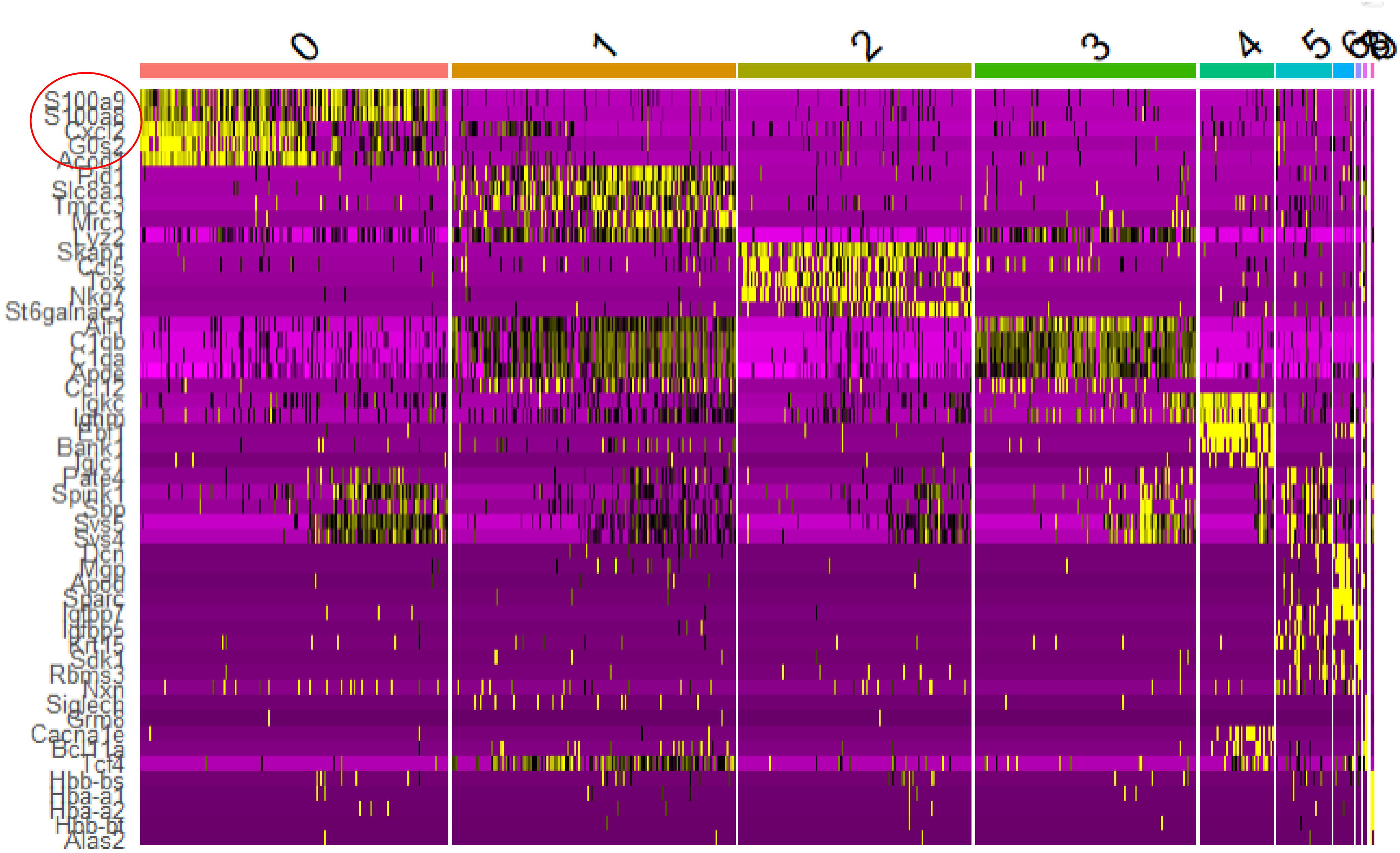
Reference: Nature. Single cell analysis of cribriform prostate cancer reveals cell intrinsic and tumor microenvironmental pathways of aggressive disease. October 2022



Cluster 4 : B cells
Cluster 2: T cells
Cluster 0, 1 ,3 y 8: Myeloid
Cluster 5: Endothelial
Cluster 6: Fibroblastos
Cluster 7: Epithelial
Cluster 9: very few genes

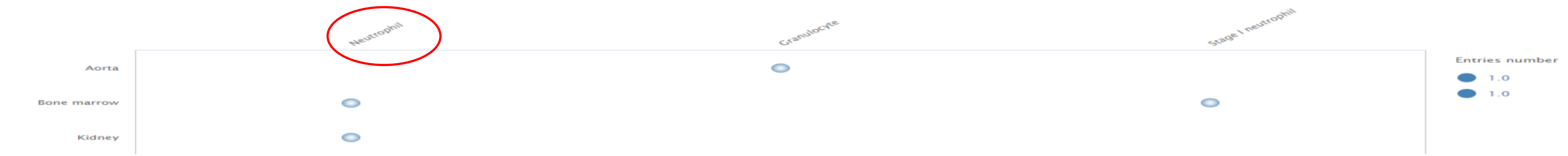


To identify the cluster 0,1,3 and 8 we use cellMarkers 2.0

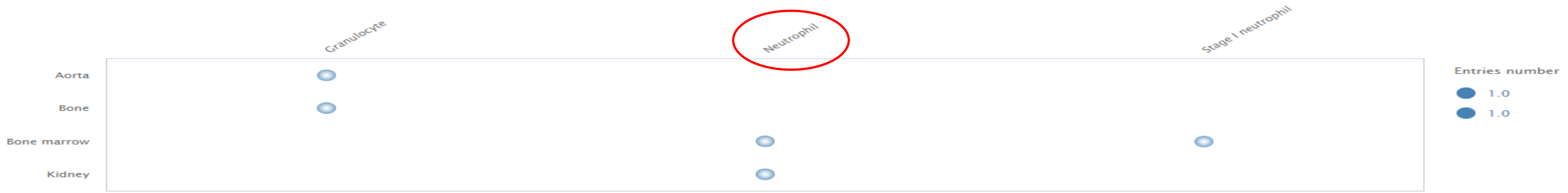


Cluster 0 markers in “CellMarker”. Here we can see what type of cells expressed this genes in different tissues.

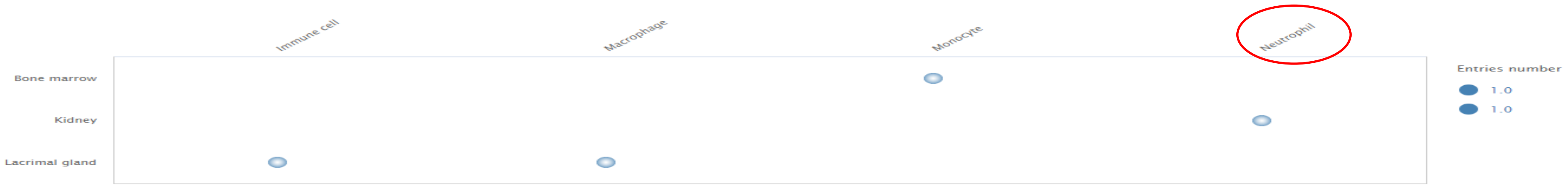
S100A9



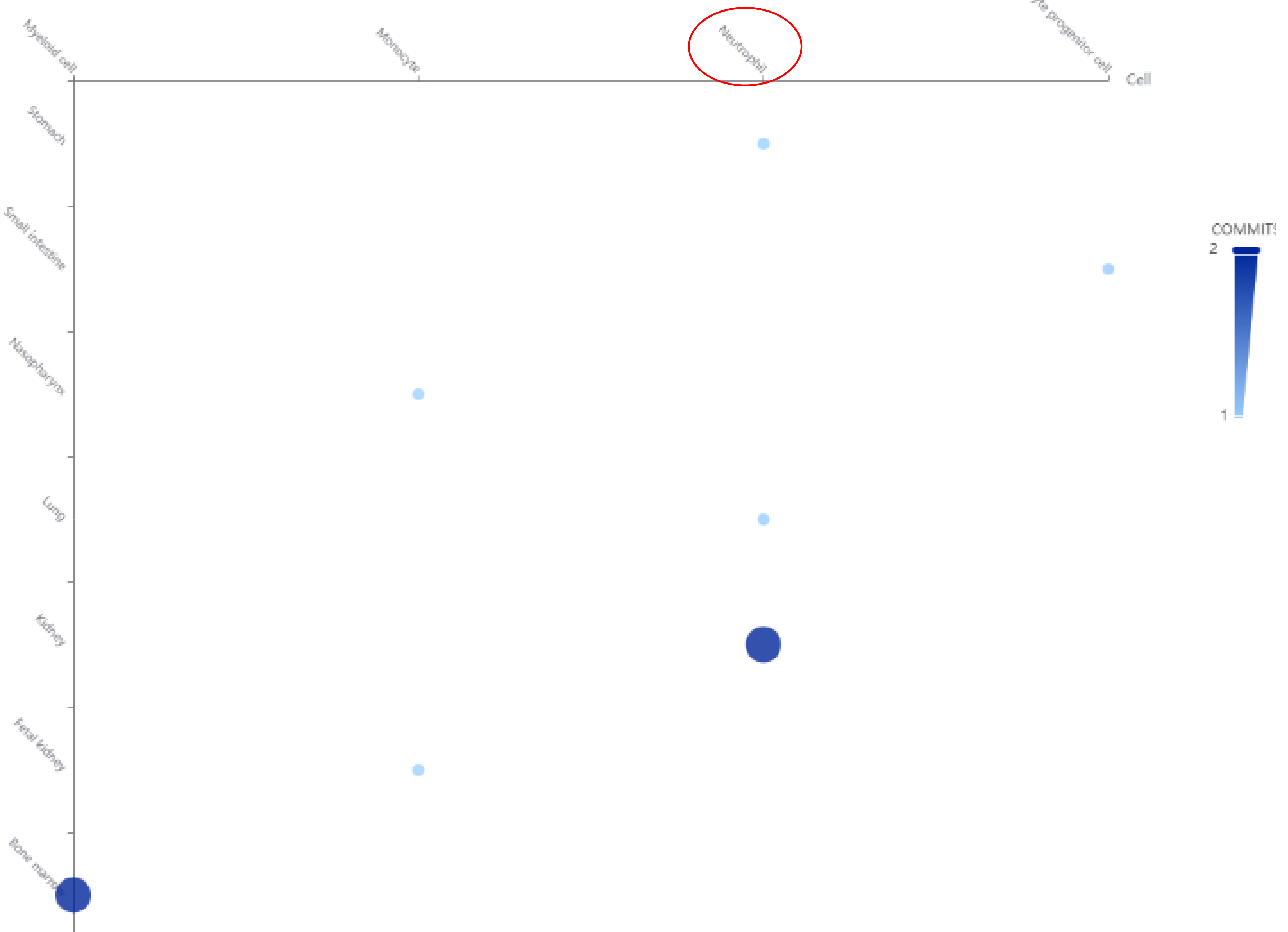
S100A8



Cxcl2



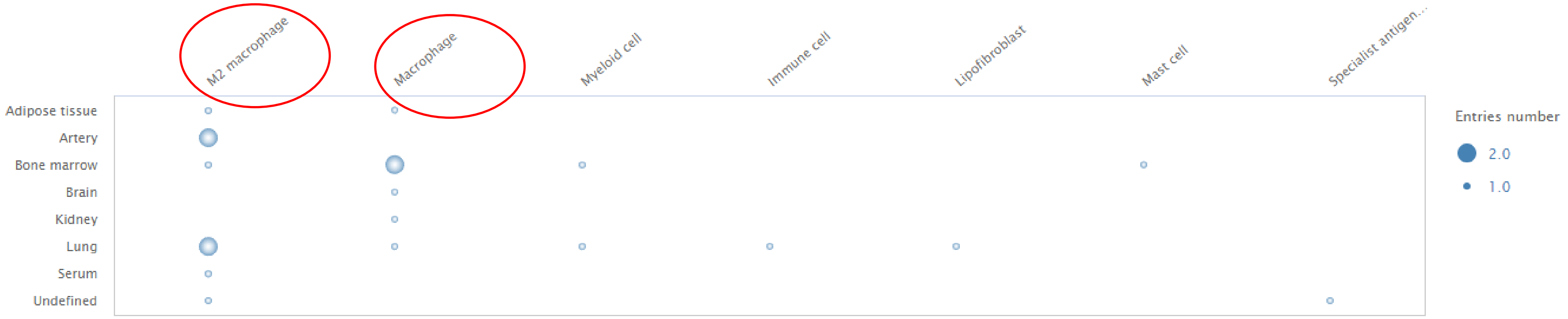
G0s2



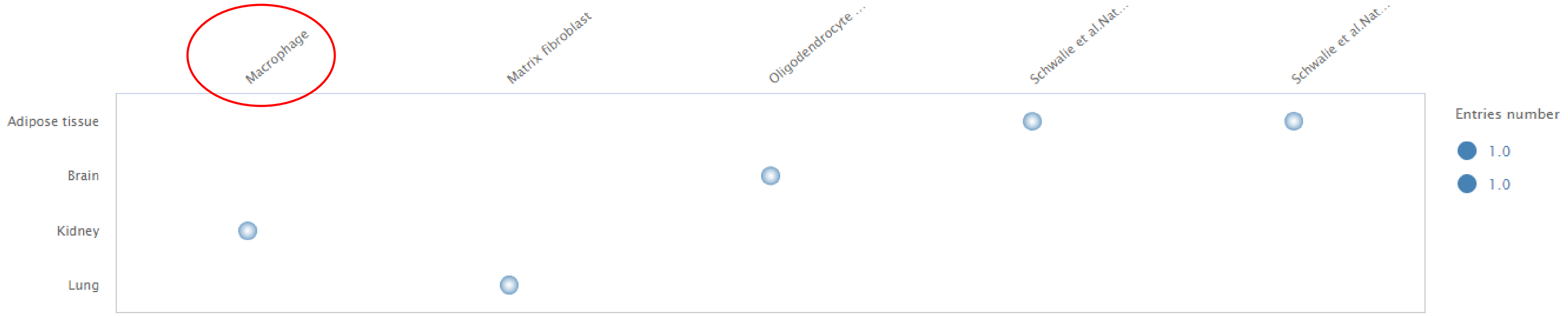
The most coincident cell type is neutrophil, so we identify the cluster 0 as neutrophils.

Cluster 1 markers in “CellMarker”

Mrc1



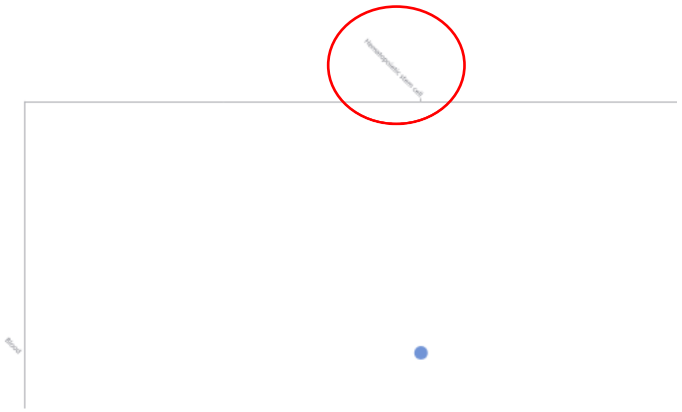
Pid1



The most coincident cell type is macrophages, so we identify the cluster 1 as macrophages.

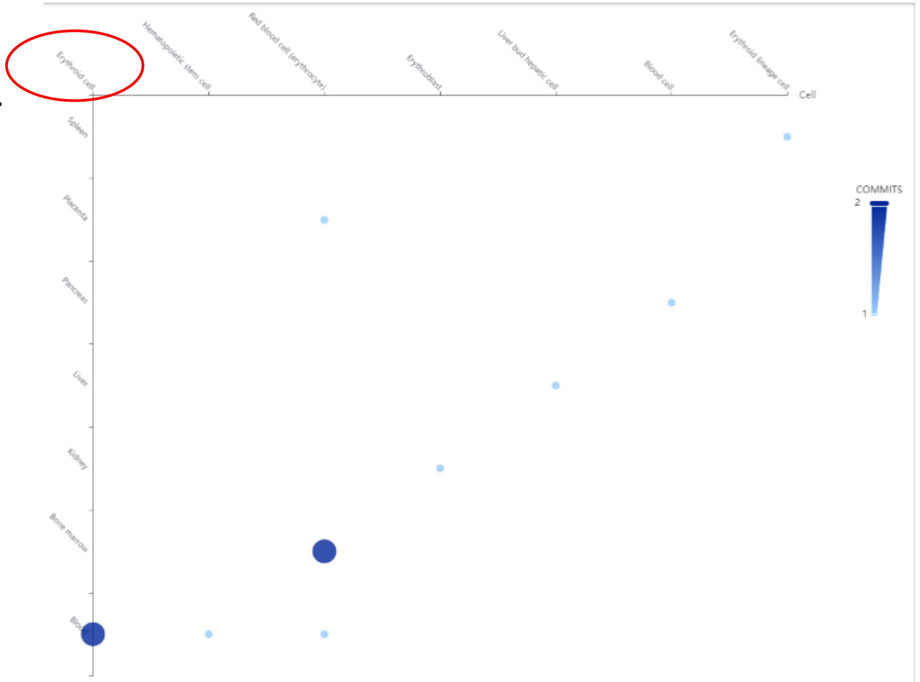
Cluster 8 markers in “CellMarker”

Hba-a1

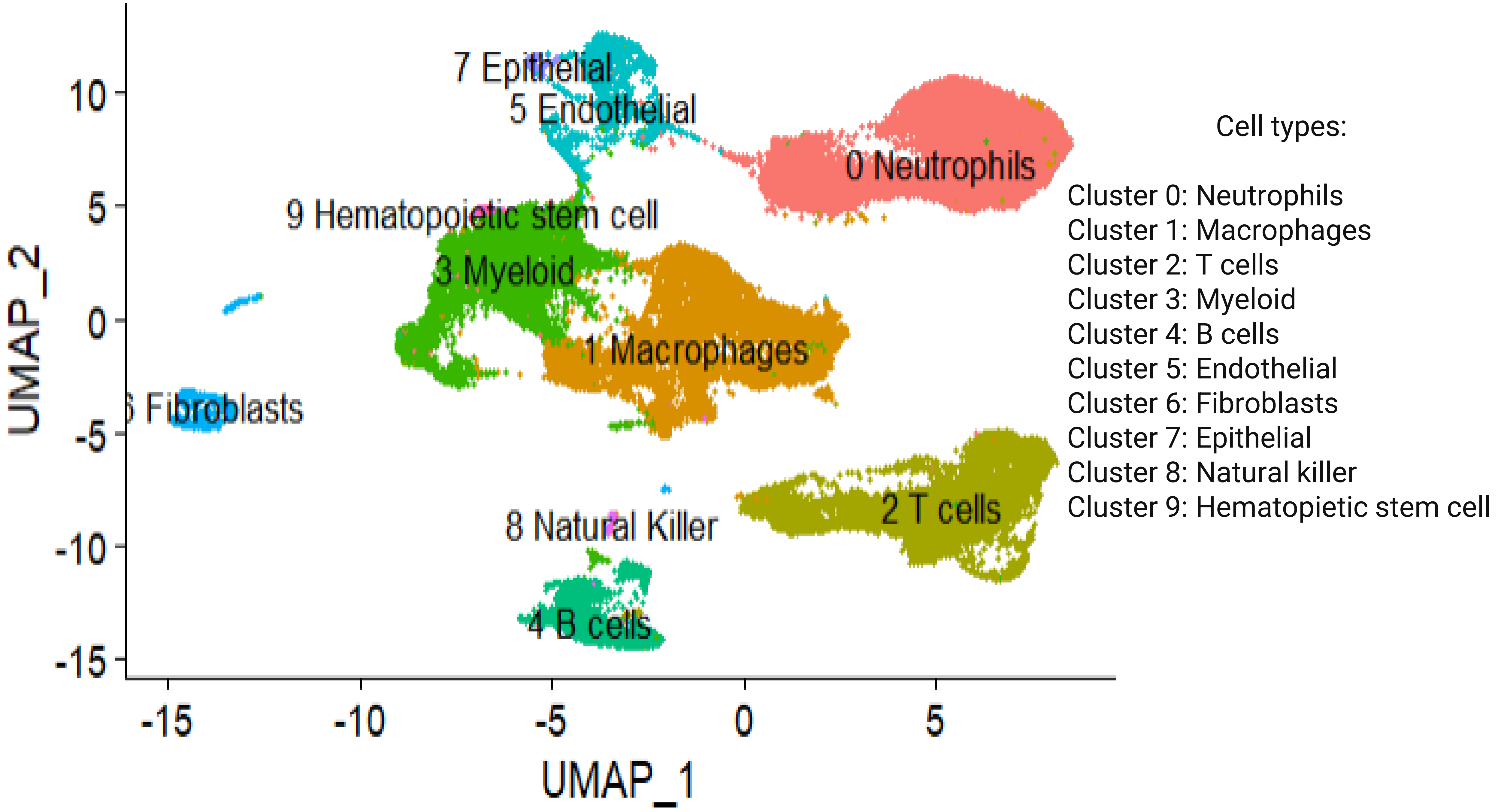


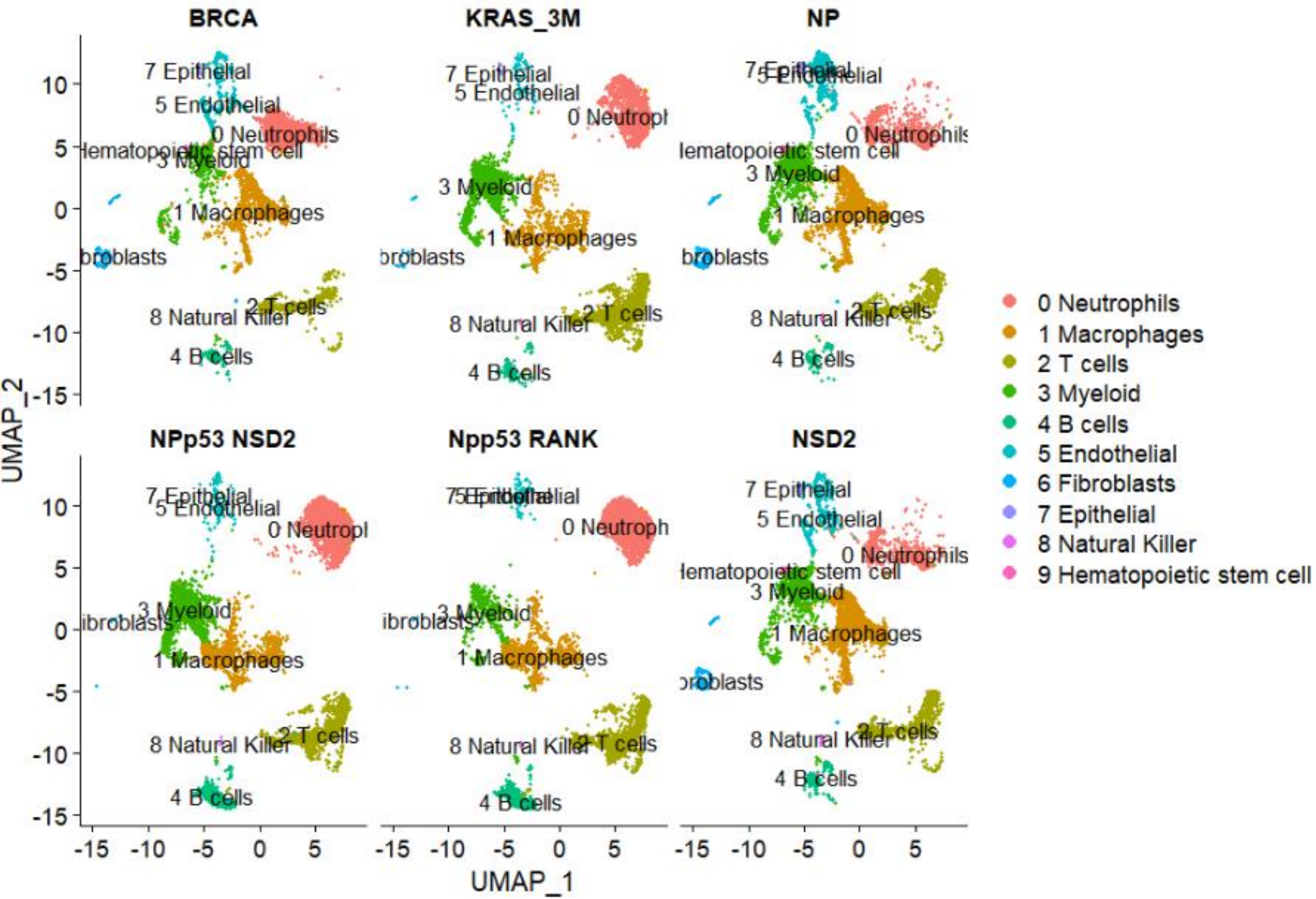
Hba-a2

Alas2



The most coincident cell type is hematopoietic stem cell and erythroid, so we identify the cluster 9 as hematopoietic stem cell.


7 Epithelial
5 Endothelial
0 Neutrophils
9 Hematopoietic stem cell
3 Myeloid
1 Macrophages
6 Fibroblasts
8 Natural Killer
2 T cells
4 B cells
10
5
0
-5
-10
-15
UMAP_2
-15
-10
-5
0
5
UMAP_1
Cell types:
Cluster 0: Neutrophils
Cluster 1: Macrophages
Cluster 2: T cells
Cluster 3: Myeloid
Cluster 4: B cells
Cluster 5: Endothelial
Cluster 6: Fibroblasts
Cluster 7: Epithelial
Cluster 8: Natural killer
Cluster 9: Hematopietic stem cell



It's the same representation than before but split by samples. In the samples KRAS_3M, NPp53 NSD2 and Npp53 RANK there is a type of neutrophil different of the other 3 samples. Maybe it would be interesting to kwon what genes are expressing differentially between this two types of neutrophils.